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Editors of *Nature Communications*
Dear Editors,

We are pleased to submit our manuscript entitled "*Cross-Body Site Microbial Interactions Influence the Human Plasma Metabolome*" for your consideration in ***Nature Communications***.

Systemic host metabolism is profoundly influenced by the human microbiome, yet most studies have focused exclusively on the gut microbiome, overlooking the potential contributions of other body-site microbial ecosystems. In this study, we present the first comprehensive analysis of how microbiomes from four major body sites, the gut, oral cavity, nasal cavity, and skin, selectively and cooperatively regulate the human plasma metabolome.

Leveraging three independent human cohorts and integrating microbiome with plasma metabolome, we demonstrate that:

(1) The microbiomes from all four body sites contribute to circulating metabolite variation, with the gut and oral microbiomes showing the most pronounced and synergistic effects.
(2) Using machine learning (GBDT) and mediation analysis, we reveal that many circulating metabolites are co-regulated by oral and gut microbiomes through cross-body site microbial interactions and directional metabolic mediation.
(3) Notably, we show that this oral-gut microbiome-metabolome axis is functionally enhanced in individuals with insulin resistance, where key metabolic pathways, including those involving aromatic and histidine amino acids, are differentially regulated.

Our findings uncover a previously underappreciated layer of host-microbiome interaction, demonstrating that cross-body site microbiomes engage in systemic and coordinated metabolic regulation. This work reframes our understanding of the microbiome's role in human physiology and suggests novel therapeutic opportunities for targeting cross-body site microbial networks in metabolic disease.

We believe that our study will be of broad interest to the readers of ***Nature Communications***, as it provides new insight into how distinct microbial communities interact across body sites to influence host systemic metabolism, and how these interactions are altered in metabolic dysfunction.

We confirm that this manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. A list of potential reviewers and conflicts of interest is provided in the submission system.

Thank you for considering our work for publication in ***Nature Communications***, and look forward to your feedback.

Yours sincerely,

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